Soils and land use

SWAT+ Documentation

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Purpose

`soils.sol` defines every soil class used in the watershed. Each class has a profile header (name, number of layers, hydrologic group, rooting depth, anion exclusion fraction, crack volume, texture string) and one row per soil layer describing depth, bulk density, available water capacity, saturated hydraulic conductivity, organic carbon, particle-size fractions, rock content, albedo, USLE K, electrical conductivity, calcium carbonate, and pH. HRUs reference these classes through `hru-data.hru`. Values feed `soil_phys_init`, `soil_nutcarb_init`, runoff, percolation, erosion, and plant uptake routines.

Source

- Reader: src/soil_db_read.f90
- Type definition: src/soil_data_module.f90, `type soil_database` (which holds `type soil_profile_db` plus an allocatable array of `type soilayer_db`)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Each soil is a multi-record block:
  - One profile record with 7 fields.
  - `nly` layer records that follow, each with 14 fields. `nly` is read from the second field of the profile record and used to allocate the layer array.
- Records are list-directed (free format). Column positions in the reference example are cosmetic indentation only; the reader does not depend on them.

Profile record (7 fields)

Read into `soildb(i)%s` (`type soil_profile_db`).

#	Field	Type	Units	Default	Description
1	snam	char(20)	none	blank	soil class name. Referenced from `hru-data.hru`
2	nly	int	none	1	number of soil layers that follow this record
3	hydgrp	char(16)	none	“A”	hydrologic soil group (A, B, C, or D)
4	zmx	real	mm	1500.0	maximum rooting depth
5	anion_excl	real	fraction	0.5	fraction of porosity from which anions are excluded
6	crk	real	fraction	0.01	crack volume potential of the soil
7	texture	char(16)	none	blank	texture descriptor string (free-form label)

Layer record (14 fields, repeated `nly` times)

Read into `soildb(i)%ly(j)` (`type soilayer_db`) for `j = 1..nly`.

#	Field	Type	Units	Default	Description
1	z	real	mm	1500.0	depth from soil surface to the bottom of the layer
2	bd	real	Mg/m^3	1.3	moist bulk density
3	awc	real	mm H2O / mm soil	0.2	available water capacity
4	k	real	mm/hr	10.0	saturated hydraulic conductivity
5	cbn	real	%	2.0	organic carbon content of the layer
6	clay	real	%	10.0	clay fraction of mineral material
7	silt	real	%	60.0	silt fraction of mineral material
8	sand	real	%	30.0	sand fraction of mineral material
9	rock	real	%	0.0	rock fragment content of the layer
10	alb	real	none	0.1	moist soil albedo
11	usle_k	real	none	0.2	USLE soil erodibility (K) factor
12	ec	real	dS/m	0.0	electrical conductivity
13	cal	real	%	0.0	CaCO3 content
14	ph	real	none	0.0	soil pH

The reader applies one fix-up after reading: if the first layer is thinner than 20 mm and either the profile has only one layer or the second layer is deeper than 20 mm, the depth of layer 1 is forced to 20 mm.

Example

Excerpt from `refdata/Ames_sub1/soils.sol`. The first soil class has four layers:

```
soils.sol Ames
   name                NLY  HYD_GRP        ZMX     ANION_EXCL      CRK       TEXTURE        DEPTH         BD
soil_01                  4        B   2000.000          0.500    0.500       L- L- L- L
                                                                                             50.00      1.260
                                                                                            150.00      1.420
                                                                                            840.00      1.500
                                                                                           2000.00      1.600
```

Notes:

- The profile record `soil_01 4 B 2000.000 0.500 0.500 "L- L- L- L"` declares 4 layers. The next four physical lines are the layer records.
- Each layer record has 14 numeric values. The `TEXTURE` token on the profile line (`L- L- L- L` here) is one space-separated token only because the reader reads `texture` as a single character field; the writer pads it with hyphen-separated horizon codes.
- Header tokens in the file (`DEPTH`, `BD`, `AWC`, `K`, `CBN`, `CLAY`, `SILT`, `SAND`, `ROCK`, `ALB`, `USLE_K`, `EC`, `CAL`, `PH`) are read past as a single skipped line. The column order is fixed by the source.

Related files

- `hru-data.hru`. Each HRU points to a soil class name in `soils.sol`.
- `nutrients.sol`. Soil chemistry initialization paired with a soil class.
- `plants.plt`. Rooting depth in `soils.sol` (`zmx`) caps plant root depth.

Purpose

`landuse.lum` is the master table of land use and management classes. Each row defines one class by name and lists string pointers to records in other files: the plant community (`plants.ini`), management schedule (`management.sch`), curve number entry (`cntable.lum`), conservation practice (`cons_practice.lum`), urban land use (`urban.urb`), overland-flow Manning's n class (`ovn_table.lum`), tile drainage (`tiledrain.str`), septic (`septic.str`), filter strip (`filterstrip.str`), grassed waterway (`grassedww.str`), and user BMP (`bmpuser.str`). HRUs reference a row in this table through `hru-data.hru`. After reading, the reader resolves every non-`null` string pointer to an integer index stored in `lum_str`; unresolved names are logged to `simulation.out` (unit 9001).

Source

- Reader: `src/landuse_read.f90`
- Type definition: `src/landuse_data_module.f90`, `type land_use_management` (instances stored in `lum`)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- One record per land use class. Records are list-directed (free format). All 14 fields are read on every row.
- Any pointer field that should be empty must be the literal string `null`. The reader treats `null` as "no link" and does not try to resolve it.

#	Field	Type	Default	Description
1	name	char(40)	blank	name of the land use and management class. Referenced from `hru-data.hru`
2	cal_group	char(40)	blank	calibration group label. Not currently used by the solver
3	plant_cov	char(40)	blank	plant community pointer into `plants.ini`. Resolved to `lum_str%plant_cov`
4	mgt_ops	char(40)	blank	management schedule pointer into `management.sch`. Resolved to `lum_str%mgt_ops`
5	cn_lu	char(40)	blank	curve number entry pointer into `cntable.lum`. Resolved to `lum_str%cn_lu`
6	cons_prac	char(40)	blank	conservation practice pointer into `cons_practice.lum`. Resolved to `lum_str%cons_prac`
7	urb_lu	char(40)	blank	urban land use pointer into `urban.urb` (e.g. `URLD`, `URHD`). Resolved during urban init
8	urb_ro	char(40)	blank	urban runoff model. `usgs_reg` for USGS regression equations, `buildup_washoff` for build-up/wash-off
9	ovn	char(40)	blank	overland-flow Manning's n class pointer into `ovn_table.lum`
10	tiledrain	char(40)	blank	tile drainage pointer into `tiledrain.str`. Resolved to `lum_str%tiledrain`
11	septic	char(40)	blank	septic pointer into `septic.str`. Resolved to `lum_str%septic`
12	fstrip	char(40)	blank	filter strip pointer into `filterstrip.str`. Resolved to `lum_str%fstrip`
13	grassww	char(40)	blank	grassed waterway pointer into `grassedww.str`. Resolved to `lum_str%grassww`
14	bmpuser	char(40)	blank	user-defined BMP pointer into `bmpuser.str`. Resolved to `lum_str%bmpuser`

The reader logs a line to `simulation.out` for every non-`null` pointer that fails to match a name in the target file. The simulation continues with the link unresolved (index zero), which means the corresponding behaviour is skipped.

Example

Excerpt from `refdata/Ames_sub1/landuse.lum`:

```
landuse.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                          cal_group          plnt_com                                  mgt
cosy_lum                           null         cosy_comm                              mgt_01
agrr_lum                           null         agrr_comm                           AGRR_rot
frst_lum                           null         frst_comm                              null
urld_lum                           null              null                              null
```

Notes:

- The header tokens written by SWAT+ Editor (`plnt_com`, `cn2`, `urban`, `ov_mann`, `tile`, `sep`, `vfs`, `grww`, `bmp`) do not match the field names in the source (`plant_cov`, `cn_lu`, `urb_lu`, `ovn`, `tiledrain`, `septic`, `fstrip`, `grassww`, `bmpuser`). The reader skips the header and uses position only.
- `urld_lum` is an urban class. Its `plant_cov` and `mgt_ops` are both `null`; the row links instead to `URLD` in `urban.urb` and selects `buildup_washoff` as the urban runoff model.
- `frst_lum` has no management schedule (`mgt_ops = null`). Forest classes typically rely on the plant community initialization alone.

Related files

- hru-data.hru. Each HRU points to one `landuse.lum` row by name.
- cntable.lum. Curve number table referenced by the `cn_lu` field. Each row holds a name and four curve numbers (hydrologic conditions A through D).
- cons_practice.lum. Conservation practice table referenced by `cons_prac`. Holds USLE P factor and maximum slope length per practice.
- ovn_table.lum. Overland-flow Manning's n table referenced by `ovn`. Holds mean, minimum, and maximum n per class.
- plants.ini. Target of `plant_cov`.
- management.sch. Target of `mgt_ops`.
- `urban.urb`. Target of `urb_lu`.
- `tiledrain.str`, `septic.str`, `filterstrip.str`, `grassedww.str`, `bmpuser.str`. Structural BMP files targeted by the last five pointer columns.

Three lookup files supply numeric parameters that landuse.lum references by name: `cntable.lum` (SCS curve numbers), `cons_practice.lum` (USLE P factor and maximum slope length), and `ovn_table.lum` (overland-flow Manning's n). Each file has the same shape: a title line, a header line, then one record per row. Rows are read with list-directed I/O, so any column after the last typed field is ignored by the solver.

cntable.lum

Curve number table. Each entry holds four SCS-CN values for hydrologic soil groups A, B, C, and D.

Source

- Reader: src/cntbl_read.f90
- Type: src/landuse_data_module.f90, `type curvenumber_table` (instances stored in `cn`)
- Sized into `db_mx%cn_lu` after the file is read.

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- One record per curve number entry. Fields are list-directed.

#	Field	Type	Default	Description
1	name	char(40)	blank	curve number entry name. Referenced from `landuse.lum` field `cn_lu`
2	cn(1)	real	30.0	curve number for hydrologic soil group A
3	cn(2)	real	55.0	curve number for hydrologic soil group B
4	cn(3)	real	70.0	curve number for hydrologic soil group C
5	cn(4)	real	77.0	curve number for hydrologic soil group D

Curve numbers are dimensionless and bounded by 0 and 100. SWAT+ Editor writes additional `description`, `treat`, and `cond_cov` columns after `cn(4)`. The reader does not consume them.

Example

Excerpt from `refdata/Ames_sub1/cntable.lum`:

```
cntable.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                      cn_a          cn_b          cn_c          cn_d  description
fal_bare               77.00000      86.00000      91.00000      94.00000  Fallow
rc_strow_g             67.00000      78.00000      85.00000      89.00000  Row_crops
wood_f                 36.00000      60.00000      73.00000      79.00000  Woods
urban                  98.00000      98.00000      98.00000      98.00000  Paved_parking_lots_roofs_drivewa
```

cons_practice.lum

Conservation practice table. Supplies the USLE support practice factor (P) and the maximum slope length used for length-slope factor calculations.

Source

- Reader: src/cons_prac_read.f90
- Type: src/landuse_data_module.f90, `type conservation_practice_table` (instances stored in `cons_prac`)
- Sized into `db_mx%cons_prac` after the file is read.

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- One record per practice. Fields are list-directed.

#	Field	Type	Default	Units	Description
1	name	char(40)	blank	-	practice name. Referenced from `landuse.lum` field `cons_prac`
2	pfac	real	1.0	-	USLE support practice factor (P). Dimensionless, 0 to 1
3	sl_len_mx	real	1.0	m	maximum slope length for the practice

SWAT+ Editor writes a `description` column after `sl_len_mx`. The reader does not consume it.

Example

Excerpt from `refdata/Ames_sub1/cons_practice.lum`:

```
cons_practice.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                     usle_p   slp_len_max  description
up_down_slope          1.00000     121.00000  Up_and_down_slope
cross_slope            0.75000     121.00000  Cross_slope_tillage
contour_farming        0.50000     121.00000  Contour_tillage
strip_contour          0.25000     121.00000  Strip_cropping_contour
ter_3-8_sodout         0.50000      76.00000  terraces_3-8%_slopes_sod-outlet
```

ovn_table.lum

Overland-flow Manning's n table. Supplies a mean, minimum, and maximum n per land use class for sheet-flow routing across the HRU.

Source

- Reader: src/overland_n_read.f90
- Type: src/landuse_data_module.f90, `type overlandflow_n_table` (instances stored in `overland_n`)
- Sized into `db_mx%ovn` after the file is read.

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- One record per class. Fields are list-directed.

#	Field	Type	Default	Units	Description
1	name	char(40)	blank	-	class name. Referenced from `landuse.lum` field `ovn`
2	ovn	real	0.5	s m^(-1/3)	mean Manning's n for overland flow
3	ovn_min	real	0.5	s m^(-1/3)	minimum Manning's n for overland flow
4	ovn_max	real	0.5	s m^(-1/3)	maximum Manning's n for overland flow

SWAT+ Editor writes a `description` column after `ovn_max`. The reader does not consume it.

Example

Excerpt from `refdata/Ames_sub1/ovn_table.lum`:

```
ovn_table.lum: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                  ovn_mean       ovn_min       ovn_max  description
convtill_nores         0.09000       0.06000       0.12000  Conventional_tillage_no_residue
convtill_res           0.19000       0.16000       0.22000  Conventional_tillage_residue
notill_2-9res          0.30000       0.17000       0.47000  No_till_2-9_t/ha_residue
forest_med             0.60000       0.50000       0.70000  Forest_medimum_good
urban_asphalt          0.01100       0.01100       0.01100  Urban_asphalt
```

Related files

- `landuse.lum`. Master land-use management table. Its `cn_lu`, `cons_prac`, and `ovn` columns name rows in these three files.
- `hru-data.hru`. Selects a `landuse.lum` row per HRU, which in turn selects rows in these three tables.